《机器人学》课程笔记

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Chapter 1

绪论

周次	讲次	教学内容（要点）
6	1	第一章：机器人的起源及发展 第二章：机器人定义及特征
7	2	第三章：机器人坐标系变换、通用旋转变换 第四章：机器人末端位姿描述 第五章：机器人坐标系及 D-H 法
8	4	第六章：机器人正运动学 第七章：机器人逆运动学
9	5	第八章：微分运动 第九章：雅可比矩阵
10	6	第十章：力雅可比 第十一章：雅可比矩阵的几何性质
11	7	第十二章：机器人刚体动力学 第十三章：机器人连杆动力学（拉格朗日）
12	8	第十四章：机器人连杆动力学（牛顿欧拉法） 第十五章：机器人连杆动力学（举例）
13	9	第十六章：关节空间的轨迹规划 第十七章：笛卡尔空间的轨迹规划
14	10	第十八章：机器人位置控制 第十九章：机器人力位混合控制
15	11	重点复习 重点复习

Table 1.1: 机器人学 I 教学日历（2026 年）

考核形式：
平时作业 (15%) + 编程作业 (20%) + 实验课 (15%) + 闭卷考试 (50%)

Chapter 2

运动学基础

2.1 机器人的自由度

Definition 2.1.1: 自由度 (Degrees of freedom, DOF)

确定机构位形所需独立参数的数目。

2.2 描述：位置、姿态与坐标系

$$\begin{array}{ccccc} \text{位姿（Location）} & = & \text{位置（Position）} & + & \text{姿态（Orientation）} \\ & & \Downarrow & & \Downarrow \\ & & 3 & & 3 \end{array}$$

Definition 2.2.1: 位置描述

$$^{A}P = \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix}$$

Definition 2.2.2: 姿态描述

旋转矩阵

$$^{A}_{B}R = \left[{}^{A}\hat{X}_B\, {}^{A}\hat{Y}_B\, {}^{A}\hat{Z}_B\right] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{31} & r_{33} \end{bmatrix}$$

对应于轴 x，y 或 z 作转角为 θ 的旋转变换，其旋转矩阵分别为：

$$R(x,\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R(y,\theta)=\begin{bmatrix}\cos\theta & 0 & \sin\theta\\ 0 & 1 & 0\\ -\sin\theta & 0 & \cos\theta\end{bmatrix}$$

$$R(z,\theta)=\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}$$

位姿（Location）＝位置（Position）＋姿态（Orientation）；位姿描述 = 位置矢量 + 旋转矩阵。表示位置时，旋转矩阵等于单位矩阵；表示姿态时，位置矢量等于零。

$$\{B\}=\{\underbrace{{}^{A}_{B}R}_{\text{旋转矩阵}},\ \underbrace{{}^{A}P_{BORG}}_{\text{原点位置矢量}}\}$$

手爪的方位——旋转矩阵 $R=\begin{bmatrix} n & o & a\end{bmatrix}$，描述手爪的姿态。

$$\{T\}=\begin{Bmatrix} n & o & a & p\end{Bmatrix}=\begin{pmatrix} n_1 & o_1 & a_1 & x\\ n_2 & o_2 & a_2 & y\\ n_3 & o_3 & a_3 & z\\ 0 & 0 & 0 & 1\end{pmatrix}$$

坐标原点位置

2.3 映射：从坐标系到坐标系的变换

2.3.1 平移映射

$\{B\}$ 与 $\{A\}$ 方位（姿态）相同，但原点不重合，用位置矢量 ${}^{A}P_{B_0}$ 表示 $\{B\}$ 的原点相对于 $\{A\}$ 的位置。

坐标平移方程：

$${}^{A}p={}^{B}p+{}^{A}p_{B_0}$$

2.3.2 旋转映射

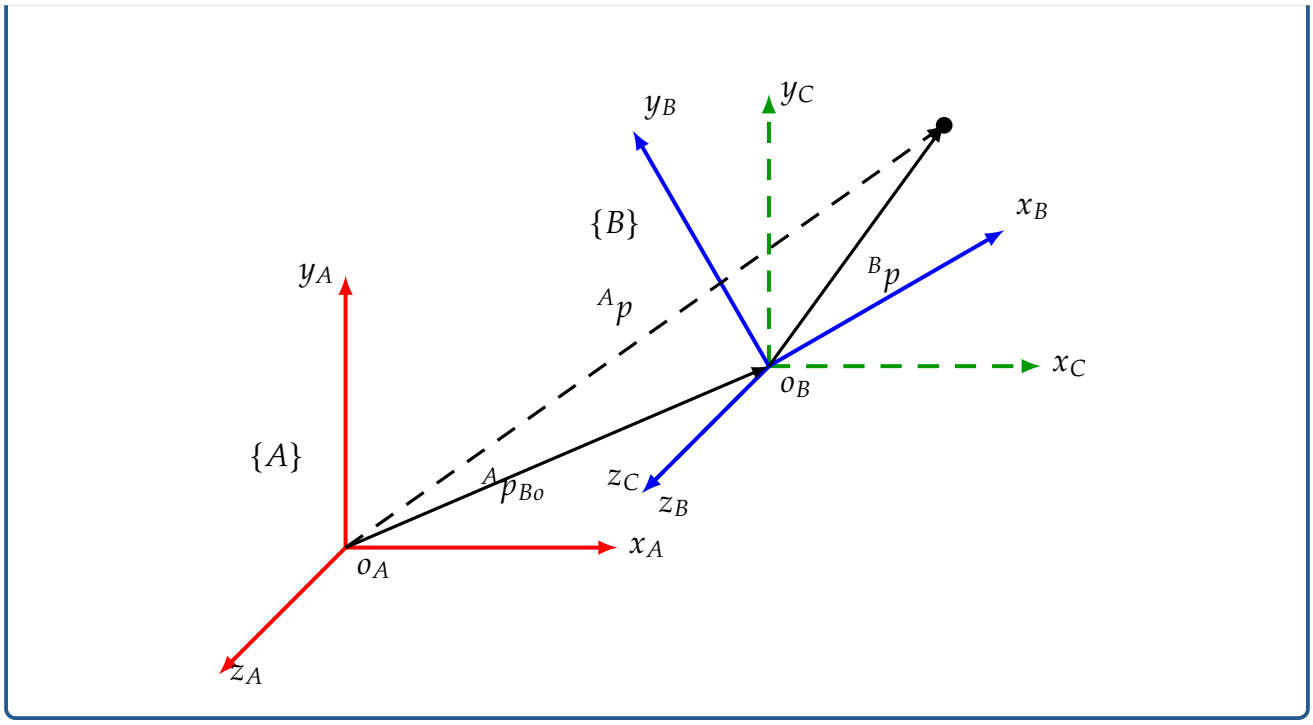
坐标旋转方程

$${}^{A}P={}^{A}_{B}R\,{}^{B}P$$

旋转的叠加：绕固定轴旋转（前乘）/ 绕当前轴旋转（后乘）

Question 1

已知坐标系 $\{B\}$ 初始位姿与 $\{A\}$ 重合，首先 $\{B\}$ 相对于 $\{A\}$ 的 z_A 轴转 30°，再沿 $\{A\}$ 的 x_A 轴移动 12 单位、沿 y_A 轴移动 6 单位。求位置矢量 ${}^{A}p_{B_0}$ 和旋转矩阵。假设点 p 在坐标系 $\{B\}$ 的描述为 ${}^{B}p=[3,7,0]^T$，求它在坐标系 $\{A\}$ 中的描述 ${}^{A}p$。



Solution:

$$
{}^Ap = {}^A_BR\,{}^Bp + {}^Ap_{B_0}
$$

$$
{}^Ap_{B_0} = \begin{bmatrix} 12 \\ 6 \\ 0 \end{bmatrix}
$$

$$
{}^A_BR = R(z, 30^\circ) = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ & 0 \\ \sin 30^\circ & \cos 30^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^Ap = {}^A_BR\,{}^Bp + {}^Ap_{B_0} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix} + \begin{bmatrix} 12 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 11.098 \\ 13.562 \\ 0 \end{bmatrix}
$$

2.3.3 一般坐标系的映射

$$
{}^Ap = {}^A_BR\,{}^Bp + {}^Ap_{B_0}
$$

$\{B\}$ 原点相对于 $\{A\}$ 的位置

$\{B\}$ 相对于 $\{A\}$ 的 (方位) 旋转矩阵

为了简化计算，将上述包含加法的仿射变换统一为单一的线性矩阵乘法，引入齐次坐标变换。定义 4×4 的齐次变换矩阵 ${}^A_BT$：

$$
{}^A_BT = \begin{bmatrix} {}^A_BR & {}^Ap_{B_0} \\ 0 & 1 \end{bmatrix}
$$

齐次坐标　　齐次坐标

$$\begin{bmatrix} {}^{A}p \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R & {}^{A}p_{B_p} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}p \\ 1 \end{bmatrix}$$

旋转矩阵　　平移矢量

$${}^{A}P = {}^{A}_{B}T\,{}^{B}P$$

Chapter 3

3.1 Random Examples

Definition 3.1.1: Limit of Sequence in $\mathbb{R}$

Let $\{s_n\}$ be a sequence in $\mathbb{R}$. We say

$$\lim_{n\to\infty} s_n = s$$

where $s \in \mathbb{R}$ if $\forall$ real numbers $\epsilon > 0$ $\exists$ natural number N such that for $n > N$

$$s - \epsilon < s_n < s + \epsilon \text{ i.e. } |s - s_n| < \epsilon$$

Question 2

Is the set $x-$axis$\backslash\{$Origin$\}$ a closed set

Solution: We have to take its complement and check whether that set is a open set i.e. if it is a union of open balls

Note:-

We will do topology in Normed Linear Space (Mainly $\mathbb{R}^n$ and occasionally $\mathbb{C}^n$)using the language of Metric Space

Claim 3.1.1 Topology

Topology is cool

Example 3.1.1 (Open Set and Close Set)

Open Set:
- ϕ
- $\bigcup_{x\in X} B_r(x)$ (Any $r > 0$ will do)
- $B_r(x)$ is open

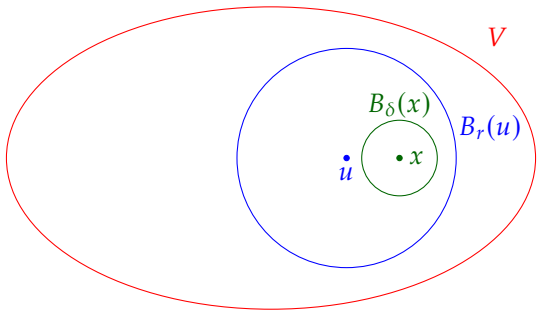
Closed Set:
- X, ϕ
- $\overline{B_r(x)}$

$x-$axis $\cup$ $y-$axis

Theorem 3.1.1

If $x \in$ open set V then $\exists\ \delta > 0$ such that $B_\delta(x) \subset V$

Proof: By openness of V, $x \in B_r(u) \subset V$



Given $x \in B_r(u) \subset V$, we want $\delta > 0$ such that $x \in B_\delta(x) \subset B_r(u) \subset V$. Let $d = d(u,x)$. Choose δ such that $d + \delta < r$ (e.g. $\delta < \frac{r-d}{2}$)

If $y \in B_\delta(x)$ we will be done by showing that $d(u,y) < r$ but

$$d(u,y) \leqslant d(u,x) + d(x,y) < d + \delta < r$$

😄

Corollary 3.1.1

By the result of the proof, we can then show...

Lenma 3.1.1

Suppose $\vec{v}_1, \ldots, \vec{v}_n \in \mathbb{R}^n$ is subspace of $\mathbb{R}^n$.

Proposition 3.1.1

$1 + 1 = 2$.

3.2 Random

Definition 3.2.1: Normed Linear Space and Norm $\|\cdot\|$

Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$). A norm on V is function $\|\cdot\|\ V \to \mathbb{R}_{\geqslant 0}$ satisfying

1. $\|x\| = 0 \iff x = 0\ \forall\ x \in V$
2. $\|\lambda x\| = |\lambda| \|x\|\ \forall\ \lambda \in \mathbb{R}$(or $\mathbb{C}$), $x \in V$
3. $\|x + y\| \leqslant \|x\| + \|y\|\ \forall\ x,y \in V$ (Triangle Inequality/Subadditivity)

And V is called a normed linear space.

• Same definition works with V a vector space over $\mathbb{C}$ (again $\|\cdot\| \to \mathbb{R}_{\geqslant 0}$) where 2 becomes $\|\lambda x\| = |\lambda| \|x\|$ $\forall\ \lambda \in \mathbb{C}$, $x \in V$, where for $\lambda = a + ib$, $|\lambda| = \sqrt{a^2 + b^2}$

Example 3.2.1 ($p-$Norm)

$V = \mathbb{R}^m$, $p \in \mathbb{R}_{\geqslant 0}$. Define for $x = (x_1, x_2, \cdots, x_m) \in \mathbb{R}^m$

$$\|x\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_m|^p\right)^{\frac{1}{p}}$$

(In school $p = 2$)

Special Case $p = 1$: $\|x\|_1 = |x_1| + |x_2| + \cdots + |x_m|$ is clearly a norm by usual triangle inequality.

Special Case $p \to \infty$ ($\mathbb{R}^m$ with $\|\cdot\|_\infty$): $\|x\|_\infty = \max\{|x_1|, |x_2|, \cdots, |x_m|\}$

For $m = 1$ these $p-$norms are nothing but $|x|$. Now exercise

Question 3

Prove that triangle inequality is true if $p \geqslant 1$ for $p-$norms. (What goes wrong for $p < 1$?)

Solution: **For Property 3 for norm-2**

When field is $\mathbb{R}$:

We have to show

$$\sum_i (x_i + y_i)^2 \leqslant \left(\sqrt{\sum_i x_i^2} + \sqrt{\sum_i y_i^2}\right)^2$$

$$\implies \sum_i (x_i^2 + 2x_i y_i + y_i^2) \leqslant \sum_i x_i^2 + 2\sqrt{\left[\sum_i x_i^2\right]\left[\sum_i y_i^2\right]} + \sum_i y_i^2$$

$$\implies \left[\sum_i x_i y_i\right]^2 \leqslant \left[\sum_i x_i^2\right]\left[\sum_i y_i^2\right]$$

So in other words prove $\langle x, y\rangle^2 \leqslant \langle x, x\rangle\langle y, y\rangle$ where

$$\langle x, y\rangle = \sum_i x_i y_i$$

Note:-

- $\|x\|^2 = \langle x, x\rangle$
- $\langle x, y\rangle = \langle y, x\rangle$
- $\langle \cdot, \cdot\rangle$ is $\mathbb{R}-$linear in each slot i.e.

$$\langle rx + x', y\rangle = r\langle x, y\rangle + \langle x', y\rangle \text{ and similarly for second slot}$$

Here in $\langle x, y\rangle$ x is in first slot and y is in second slot.

Now the statement is just the Cauchy-Schwartz Inequality. For proof

$$\langle x, y\rangle^2 \leqslant \langle x, x\rangle\langle y, y\rangle$$

expand everything of $\langle x - \lambda y, x - \lambda y \rangle$ which is going to give a quadratic equation in variable λ

$$\begin{aligned}
\langle x - \lambda y, x - \lambda y \rangle &= \langle x, x - \lambda y \rangle - \lambda \langle y, x - \lambda y \rangle \\
&= \langle x, x \rangle - \lambda \langle x, y \rangle - \lambda \langle y, x \rangle + \lambda^2 \langle y, y \rangle \\
&= \langle x, x \rangle - 2\lambda \langle x, y \rangle + \lambda^2 \langle y, y \rangle
\end{aligned}$$

Now unless $x = \lambda y$ we have $\langle x - \lambda y, x - \lambda y \rangle > 0$ Hence the quadratic equation has no root therefore the discriminant is greater than zero.

When field is $\mathbb{C}$:

Modify the definition by

$$\langle x, y \rangle = \sum_i \overline{x_i} y_i$$

Then we still have $\langle x, x \rangle \geqslant 0$

3.3 Algorithms

Algorithm 1: what

Input: This is some input

Output: This is some output

```
/* This is a comment */

some code here;
x ← 0;
y ← 0;
if x > 5 then
    x is greater than 5 ;                    // This is also a comment
else
    x is less than or equal to 5;
end
foreach y in 0..5 do
    y ← y + 1;
end
for y in 0..5 do
    y ← y − 1;
end
while x > 5 do
    x ← x − 1;
end
return Return something here;
```